CS 3244

Machine Learning

Week 13 – Exam Format and Revision

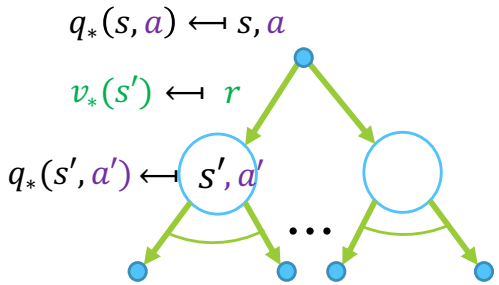
13 post

Kan Min-Yen

Background Photo credits: Rafiq Mirza, Luan Ahn, Rosmarie Voegtli @ Flickr

Recap and Forecast

- Understand the basic anatomy and terminology in reinforcement learning
  - Policy, reward, action, observation and (environment, agent) model
- Introduce the Markov property and use this to define a Markov decision process (MDP)
- Understand the Bellman equation as a recursive form for calculating future rewards.



$$q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \max_{a'} q_*(s', a')$$

Learning Outcomes for this week:

- Let's get FAT (i.e., Fairness, Accountability and Transparency)
- Have awareness of ML in the larger context, especially how it can affect people's lives other than ourselves;
- Understand the systematic shortcomings of problems with machine learning, in its stages of:
  - Data gathering;
  - Preparation;
  - Model learning.

Epilogue

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Course Learning Outcomes

1. Understand the basic concepts of machine learning.

2. Apply an appropriate machine learning algorithm for a given problem.

3. Evaluate the performance of a machine learning solution.

4. Use a machine learning tool to carry out machine learning experiments.

Secondary Learning Objectives

In addition to learning about Machine Learning, you have picked up skills that help with your future career.

- **Python** – one of the easiest and more straightforward programming languages to use.
- **Jupyter Notebook / Google Colab** – an intuitive way to mix visualization and code, and get access to state-of-the-art resources.
- **Practical Project** – learn best by applying what you know on a project of your design and communicating it with others.

What we didn't cover

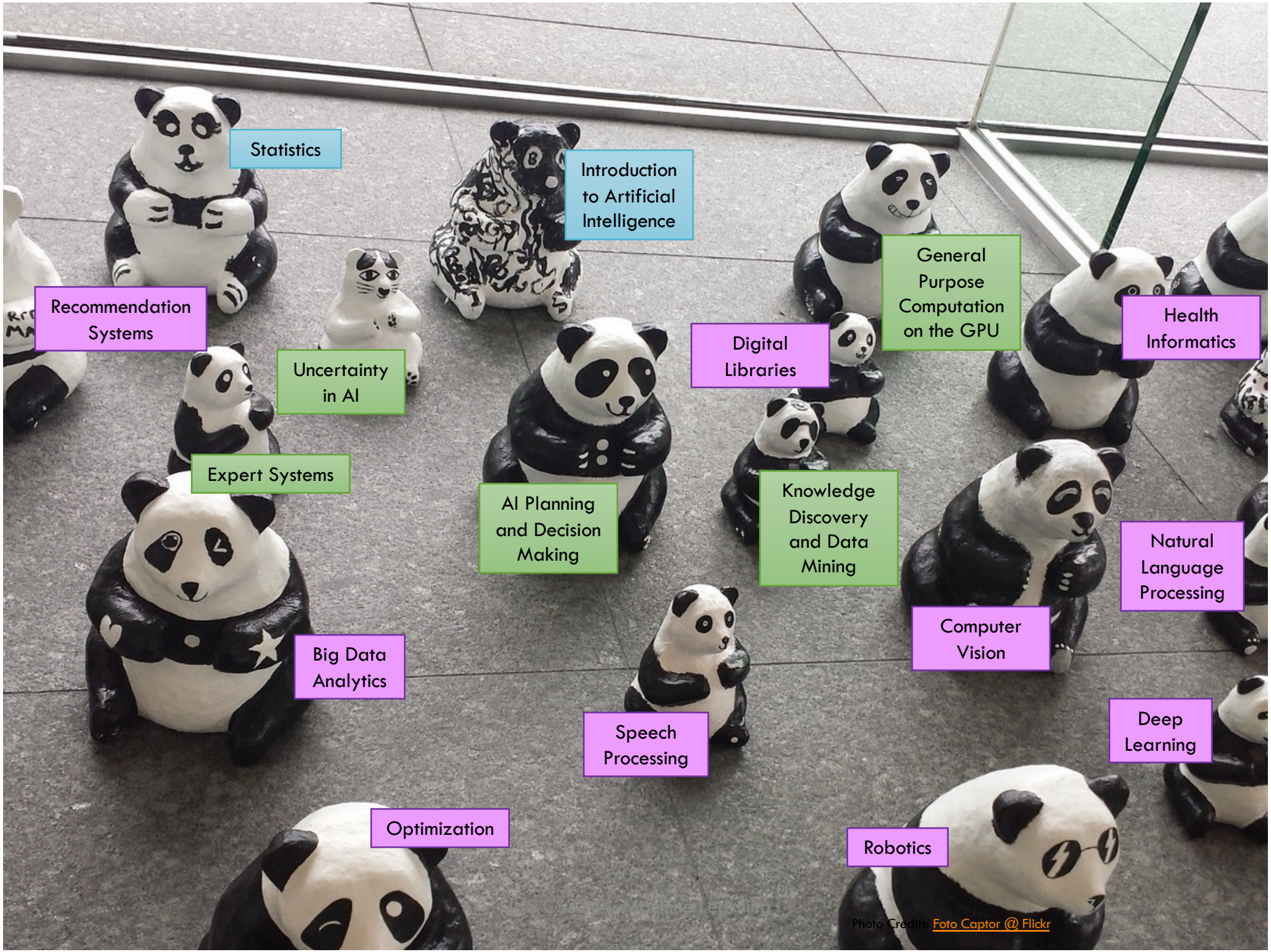
Many things, but notably:

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

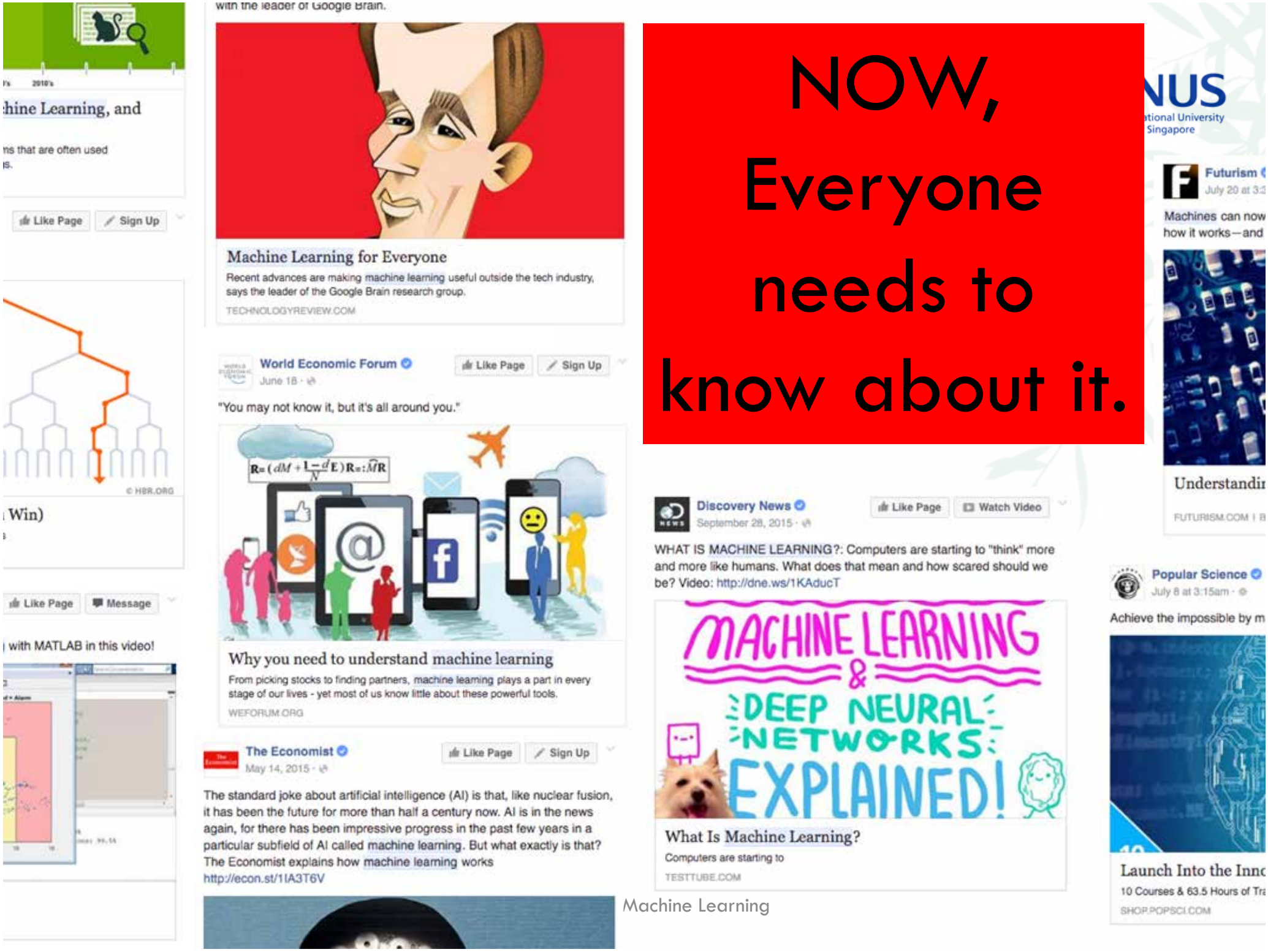
Bayesian Models!

But we did touch upon these in tutorials.

You'll see it again in some other NUS machine learning classes.


Statistics
Introduction to Artificial Intelligence
Recommendation Systems
Uncertainty in AI
Expert Systems
AI Planning and Decision Making
Digital Libraries
General Purpose Computation on the GPU
Health Informatics
Knowledge Discovery and Data Mining
Natural Language Processing
Computer Vision
Big Data Analytics
Speech Processing
Deep Learning
Optimization
Robotics
Photo Credits: Foto Captor @ Flickr

NOW, Everyone needs to know about it.



… but perhaps only a few understand it.

arr

Goodbye and Acknowledgments

Final Assessment

25 Nov (Mon) 13:00-15:00.

Venue: MPSH2-B.

Open Book, physical notes & calculator.
No electronic devices (sorry).

Stay in touch with our CS3244 alumni and general interest group on Facebook (search for "cs3244")

Thanks for joining us for the journey!

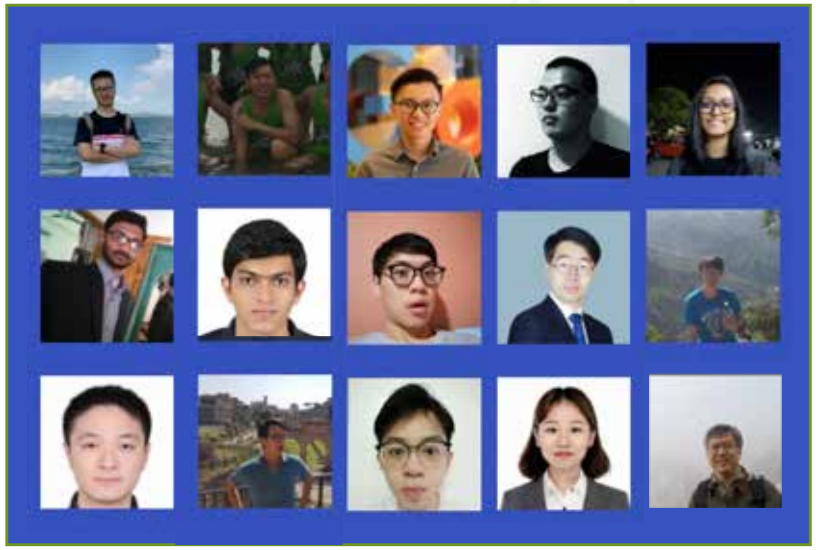


Photo Credits: People's Online Daily (English Version)

Exam Format

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Exam Format

Open Book

But no electronic devices, sorry!

~ 40% will be MCQ/MRQ

Topics not necessarily in order

100 points
120 minutes

Min's past finals:
Sem1 of 2017/18
Sem1 of 2016/17

INSTRUCTIONS TO CANDIDATES

1. Please write your Student Number below. Do not write your name.
2. This assessment paper contains FIVE (5) questions and comprises EIGHT (8) printed pages, including this page. Some questions have multiple parts.
3. It is suggested that you limit your response length to the space in the boxes provided.
4. You may use the backs of the pages as scratch paper, as they *will* be disregarded, unless specifically noted by you.
5. This is an OPEN BOOK assessment. You may consult books and any other printed or handwritten materials for this test.
6. No electronic devices, except for calculators, are permitted.
7. You may use pencil or other erasable medium in answering this paper.
8. The questions are presented no particular order, and specifically not by their perceived difficulty or estimated time to answer. You may want to do the questions out of order.

STUDENT NO: _____

Specimen. Not the actual final

This portion is for examiner's use only

Question	Q1	Q2	Q3	Q4	Q5	Total
Max	**20**	**18**	**12**	**24**	**26**	**100**
Marks						

Exam Topics

All topics covered in pre- and post-lecture (through slides).

But note that some of the NUS digital library's previous exam questions are from a different lecturer.

1 programming question but where the suitable function call prototypes, and library information are given to you.

- Pseudocode ok!

Not responsible for sections that we didn't cover

- If in doubt, ask on the forum.

Exam Topic Distribution

Emphasize on second half of semester.

2-3 questions may have calculation.

More time consuming but easy and straightforward.

Don't forget your calculator!

Others are short answer and thinking essay questions.

Week 1 12 Aug	Administrivia and Paradigms of Learning
Week 2 19 Aug	Concept Learning
Week 3 26 Aug	Naïve Bayes and k-Nearest Neighbors *T01: Paradigms of Learning*
Week 4 2 Sep	Linear Classifiers and Logistic Regression *T02: Concept Learning, Naïve Bayes and k-Nearest Neighbors*
Week 5 9 Sep	Bias and Variance and Overfitting *T03: Linear Classifiers and Logistic Regression*
Week 6 16 Sep	Regularization and Validation *T04: Bias, Variance and Overfitting*
Week 8 7 Oct	Neural Networks *T05: Regularization and Validation*
Week 9 14 Oct	Deep Learning *T06: Neural Nets*
Week 10 21 Oct	Decision Trees and Ensembles *T07: Deep Learning*
Week 11 28 Oct	Unsupervised ML: k Means and Expectation Maximization *T08: Decision Trees and Ensembles*
Week 12 4 Nov	Reinforcement ML *T09: Unsupervised Learning*
Week 13 11 Nov	Machine Learning Ethics *T10: Reinforcement ML*

1st half: about 30%

2nd half: about 70%

Pre-Midterm Revision

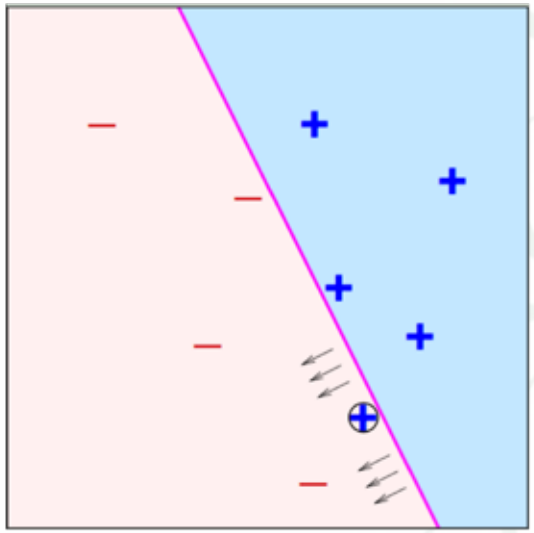
CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Week 1 – Paradigms of Learning

Learning is used when

1. A pattern exists
2. We cannot pin it down mathematically
3. We have data on it



Example: PLA

Focus on supervised learning

- Unknown target function
  $$y = f(x)$$
- Data set
  $(x^{(1)}, y^{(1)}), \ldots, (x^{(m)}, y^{(m)})$
- Learning algorithm picks h to approximate f from a hypothesis set $\mathcal{H}$.

Learning an unknown function?

- Impossible. The function can assume any value outside the data we have
- So what now?

Week 2 – Concept Learning

Concept learning as search through $\mathcal{H}$, possibly of infinite size.

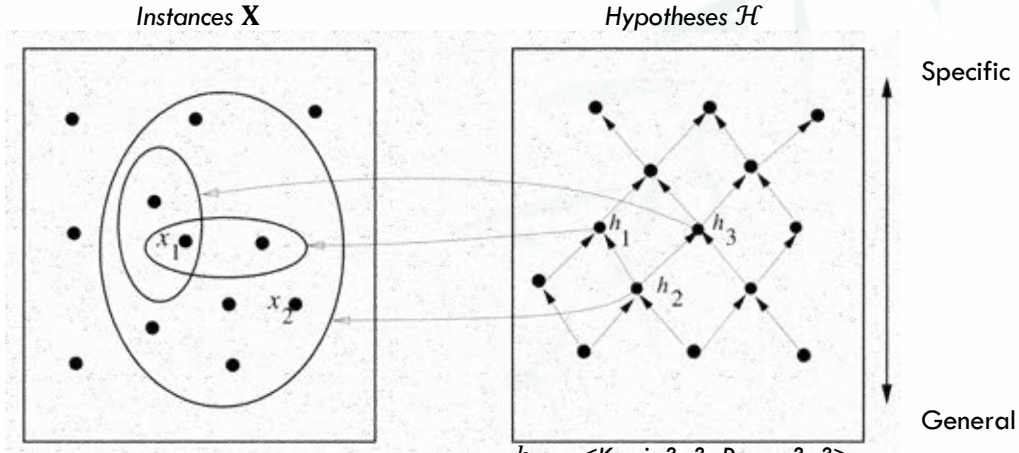
Learn concept learning through the hypothesis space of conjunctive hypotheses:

Drink	Milk	Sugar	Iced	Strength	For Here
$\langle ?,$	$Si,$	$?,$	$Peng,$	$?,$	$Lim\rangle$

Futility of Bias-Free learning: All $\mathcal{A}$ will necessarily need to be biased in order to make predictions on future data.

Candidate elimination algorithm

- Boundaries S and G characterize learner's uncertainty
- Active learner can generate informative queries



$\mathbf{x}^{(1)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>
$\mathbf{x}^{(2)}$ = <Kopi, Si, Ga Dai, –, Por, Lim>

h_1 = <Kopi, ?, ?, Peng, ?, ?>
h_2 = <Kopi, ?, ?, ?, ?, ?>
h_3 = <Kopi, ?, ?, ?, Por, ?>

Week 3 – Naïve Bayes & k NN

Naïve Bayes:

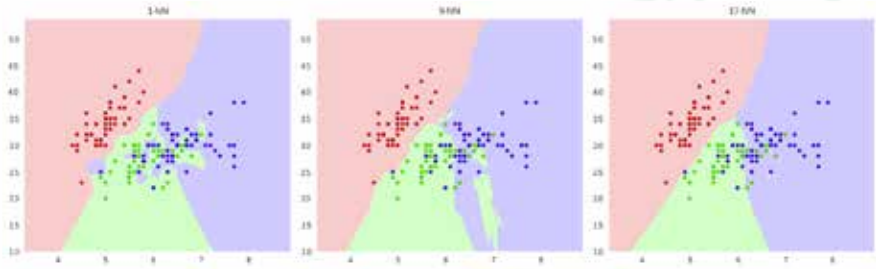
- Naïve: assumes independence between input features
- Bayes: estimates posterior from prior, marginals and likelihoods

$$P(y|x) = P(x_1|y) \times \cdots \times P(x_n|y) \times P(y)$$

- Smoothing caters for unseen data

k NN:

- Instance-based non-parametric
- Votes from nearest neighbor
- Larger k to smooth decision boundaries



- Curse of Dimensionality: works poorly for high dimensional data

Week 4 – Linear Models

Linear models use the signal:

$$s = \theta^\top x$$

- Classification: $h(x) = \text{sign}(\theta^\top x)$
- Regression: $h(x) = \theta^\top x$

Linear Regression

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

Closed form normal equation:

$$\theta = (X^\top X)^{-1} X^\top y$$

Nonlinear transformation

- $\theta^\top x$ is linear in θ
- Any $x \xrightarrow{\Phi} z$ preserves this linearity
- E.g., $(x_1, x_2) \xrightarrow{\Phi} (x_1^2, x_2^2)$

Cost measures

Application specific, user should specify

False accepts and rejects may differ in badness

Noisy targets

$$y = f(x) \quad \rightarrow \quad y \sim P(y|x)$$

Week 4 – Logistic Regression

Logistic Regression

- Model probability of the observed data
- Optimize for (maximum) likelihood
- Equivalent to minimizing cross entropy
- To solve: No closed form, possible since logistic function is convex (only one minimum)

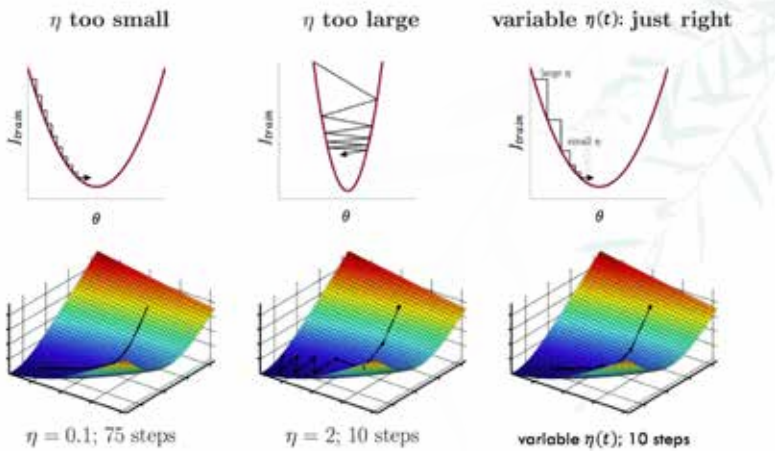
Logistic Regression

- Probability of a binary event

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + e^{-y^{(j)}\theta^{\top}x^{(j)}})$$

Gradient Descent

- Take steps in opposite the gradient of the cost function.
- SGD: the average pointwise direction is equivalent to the gradient $-\nabla J$



Stochastic Gradient Descent

- Faster, approximate method
- Think PLA + GD

Week 5 – Bias/Variance and Overfitting

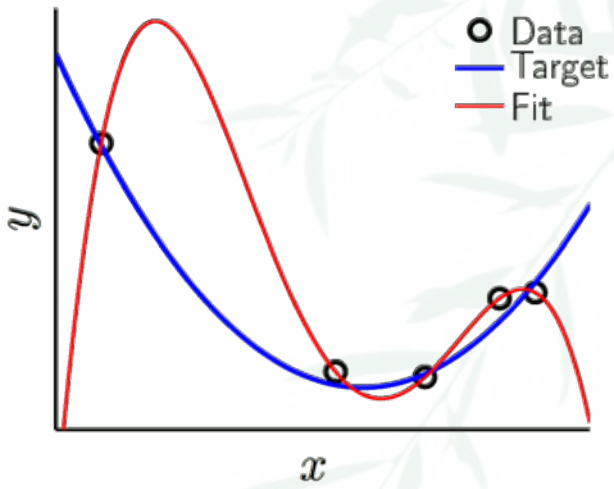
Bias-Variance Tradeoff

Bias: How well $\mathcal{H}$ can approximate f overall

Variance: How well we can zoom in on a good $h \in \mathcal{H}$

Match the 'model complexity' to the data resources, not to the target complexity

Overfitting: Fitting the data more than is warranted



Two causes: fitting stochastic + deterministic noise

Bias $\equiv$ deterministic noise

Post-Midterm Revision

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

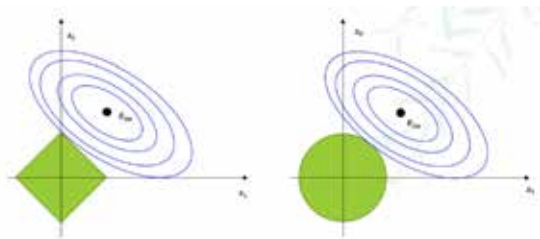
Week 6 – Regularization and Validation

1. Regularization

Give up modeling a subset of of $\mathcal{H}$ to gain better variance.

Minimize $J_{aug}(\theta) = J_{train}(\theta) + \frac{\lambda}{m}\Omega(h)$

Choosing a regularizer

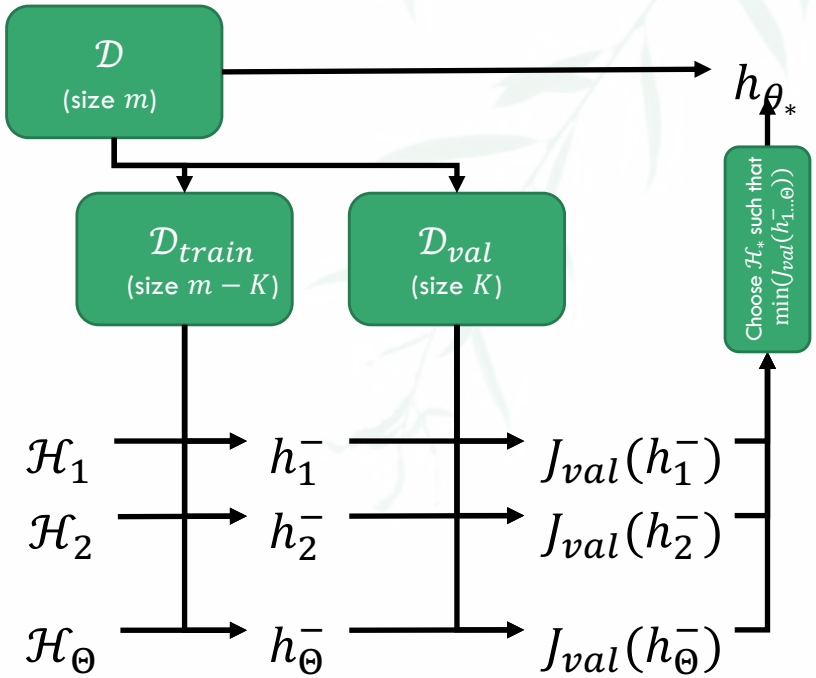


Common $\Omega(h)$: weight decay

Choose λ by validation …

2. (Cross) Validation

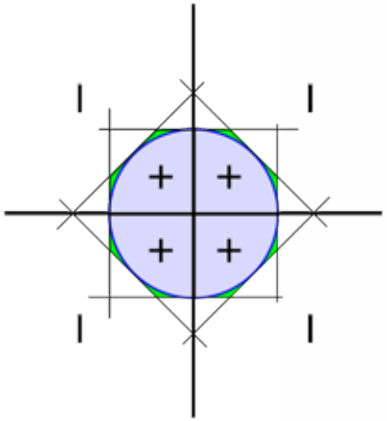
$$J_{test}(h) \approx J_{test}(h^{-}) \approx J_{val}(h^{-})$$



Week 8 – Neural Networks

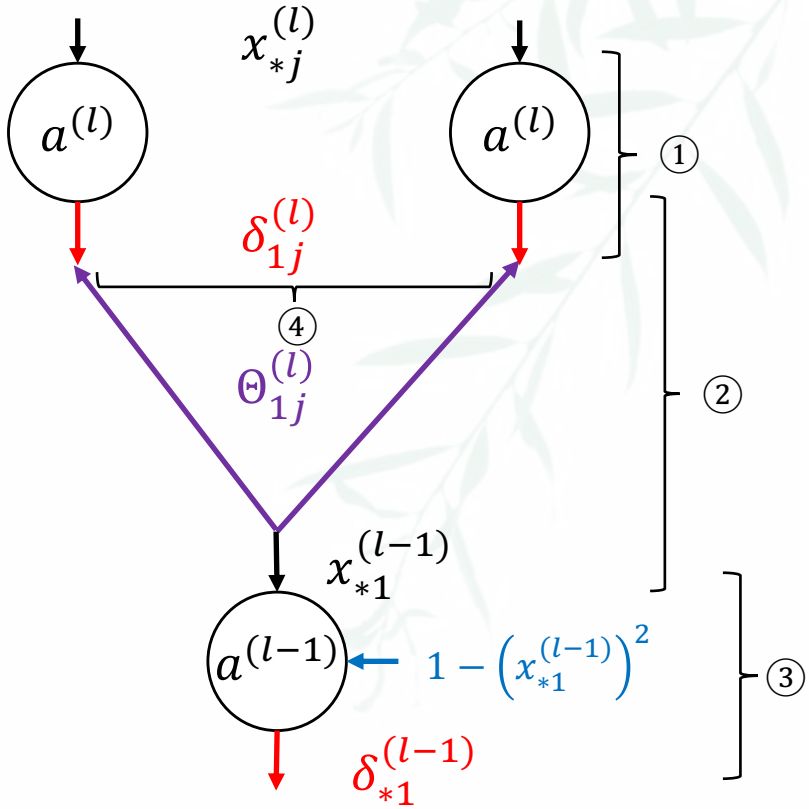
Neural Network

Logical combination
of linear units



$$x_j^{(l)} = g\left(\sum_{i=0}^{d^{(i-1)}} \Theta_{ij}^{(l)} x_i^{(l-1)}\right)$$

where $g(s) = \tanh(s)$



Backprop

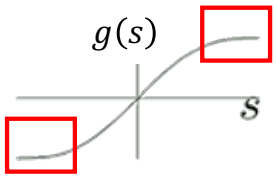
$$\Delta\Theta_{ij}^{(l)} = -\eta\, x_i^{(l-1)} \delta_j^{(l)}$$

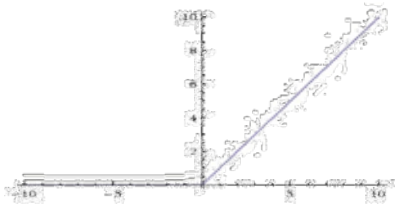
where $\delta_i^{(l-1)} = \left(1-\left(x_i^{(l-1)}\right)^2\right)\sum_{j=1}^{d^{(l)}} \Theta_{ij}^{(d)} \delta_j^{(l)}$

Week 9 – Deep Learning

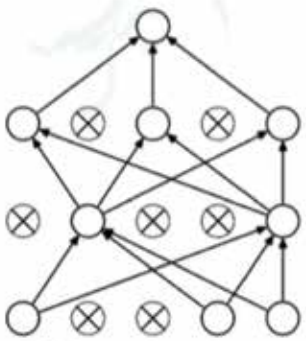
Deep Learning
≡ Stacked NNs
3 important innovations:

1. ReLU – Vanishing Gradient





2. Initialization – Pre-training via autoencoding
3. Dropout – Regularization



(b) After applying dropout.

Two basic variants to regularize toward useful $\mathcal{H}$:

1. <u>Recursive NN</u>: reuse same Θ for multiple time steps in a sequence;
2. <u>Convolutional NN</u>: reuse same Θ for different local regions in an input.

Transfer Learning: Reuse network tuned for a previous task on a current task (*cf* word embeddings).

Week 10 – Decision Trees and Ensembling

Decision Trees

Expressive: can overfit to achieve $J_{train} = 0$

Need pruning to combat variance

Bootstrap & Bagging

Sample with replacement to generate new $\tilde{\mathcal{D}}_t$.

With each train a h_t, apply uniform voting.

Variance lowered through averaging

Boosting

Make a strong classifier from simple weak classifiers.

Reweight $\tilde{\mathcal{D}}_t$ to account for easy / hard examples.

Lower bias via simple hypotheses.

Random Forest

Combine Bagging with DTL.

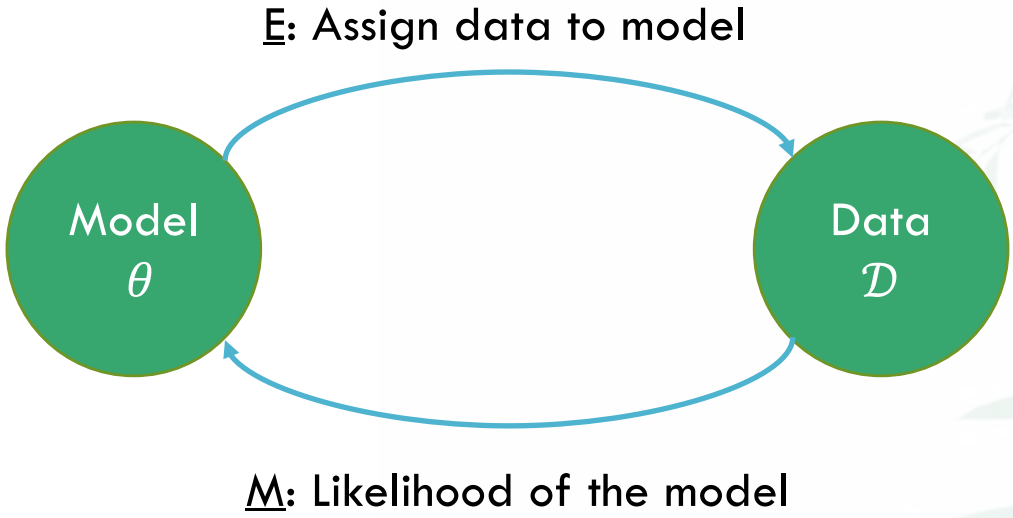
Introduce more randomness through feature projection.

Each h_t tuned to some subset of the features.

Bottom Line: Diversity is good

Week 11 – K means and Expectation Maximization

The EM Algorithm



K means as as simplified form of the EM meta-algorithm, where cluster assignments are hard, and using an $\mathcal{L}^2$ norm.

Many instances of EM out there, instantiated for different classes of problems; e.g., "forward-backward", "inside-outside"

Week 12 – Reinforcement Learning

Understand the basic anatomy and terminology in reinforcement learning

- Policy, reward, action, observation and (environment, agent) model

Introduce the Markov property:

$$P(s_{t-1}|s_t) = P(s_{t+1}|s_1, \dots, s_t)$$

Use it to define:

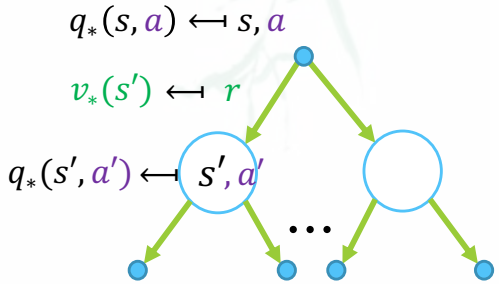
- Markov Chain (MC; $\langle \mathcal{S}, \mathcal{P} \rangle$)
- Markov Reward Process (MRP; $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$)
- Markov Decision Process (MDP; $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}^a, \mathcal{R}, \gamma \rangle$)

Understand the Bellman equations as a recursive form for calculating future rewards.

- Expectation Equations; linear
- Optimal Equations; max, non-linear

View as an interleaved tree of states and actions:

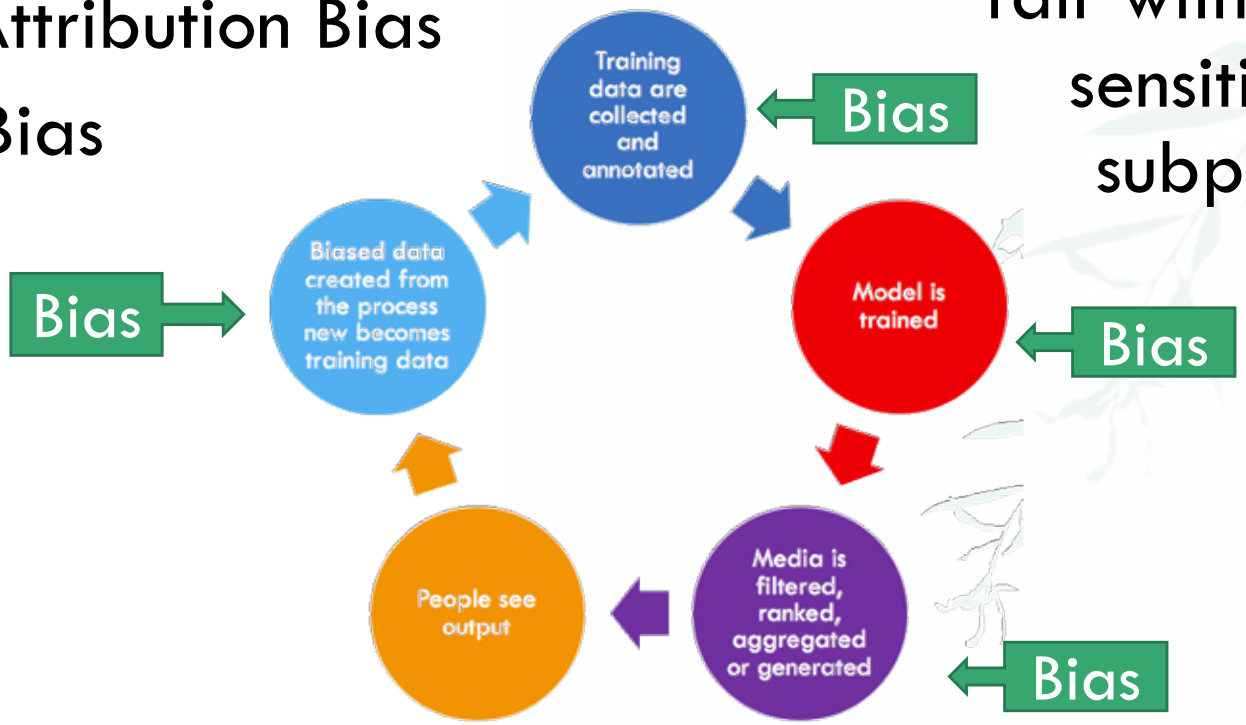
e.g. $q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \max_{a'} q_*(s',a')$



Week 13 – Ethics

Bias

- Reporting Bias
- Automation Bias
- Selection Bias
- Group Attribution Bias
- Implicit Bias

Ethics

- Remove or control for sensitive fields in training
- Validate outcomes are fair with respect to sensitive fields / subpopulations.